Lewis McGuire
Glasgow | 07401 118084 | mcguirelewis6@gmail.com
LinkedIn: linkedin.com/in/lewis-mcguire-dundee-uni

Personal Profile

Fourth-year Computer Science student at the University of Dundee, passionate about low-level programming, real-time software, and embedded systems. I enjoy working close to hardware and developing reliable and efficient software, drawing on my experience from projects involving microcontrollers and CPU design. My exchange abroad has strengthened my adaptability and resilience, while my leadership roles have enhanced my communication and collaboration skills. I am seeking a graduate software engineering role to further develop my technical skills and knowledge through engineering projects with real-world impact.

Work Experience

Summer Placement Software Engineer
Leonardo UK, Edinburgh | June – Aug 2026

- Worked in an Agile team environment, continuously learning from review feedback and applying it to my future work, improving its quality and maintainability.
- Contributed to a large existing codebase using C++ and MATLAB, quickly learning new tools and technologies by analysing the existing code.
- Gained valuable experience using Atlassian tools: using version control with Bitbucket, creating and tracking tickets in Jira, and creating design documentation in Confluence.

Education

BSc (Hons) in Computer Science
University of Dundee | Expected Graduation: June 2027

- Maintaining an A average (on track for First-Class Honours)
- Duff and Garland Bursary, Level 3 Class Prize Computer Science

Non-Award Exchange Programme (6 Months)
Australian National University | 2025

- Built independence and resilience by adapting to a new country and academic system
- Navigated a demanding and challenging workload whilst maintaining strong grades

---

Technical Skills

- Programming Languages**:** C++, MATLAB, Python, Erlang, Java, C, Assembly, JavaScript
- Embedded & Systems: Finite State Machines, Interrupts, Microcontrollers, CPU Design
- Tools: Git, Atlassian Suite, VS Code, Linux (Ubuntu), IntelliJ
- Testing: Unit, Integration, Functional, Usability
- Other Technologies: PostgreSQL, HTML & CSS, Flask, Firebase, MongoDB

Projects

Honours Project (In Progress): New Parallel Patterns for the Skel Library (Erlang)

- Designing skeleton algorithms and benchmarks to simplify concurrent programming.
- Evaluating solutions based on performance, scalability and code conciseness.

Industrial Team Project (Python, Web, Databases)

- Took on the role of Scrum Master, leading a team of six using Agile methodology.
- Ensured team cohesion by facilitating cross-team updates from the data parsing, analysis, and UX subgroups as well as utilising a shared Kanban board.
- Demonstrated progress updates to the client and incorporated their feedback, adapting plans and reprioritising backlog items for each sprint.

BBC micro:bit Light Show & Mini-Game (ARM Assembly)

- Extended a dynamic LED light show into a responsive real-time mini-game running on constrained microcontroller hardware, requiring register-level management.
- Structured the program around a finite-state machine to manage animations, state transitions, and game logic, using jump-tables to keep the code modular and scalable.
- Used timers and GPIO interrupts to keep animations non-blocking, handle input responsively, and reliably detect short and long button presses with debouncing.

CPU Microarchitecture & ISA Design (Digital)

- Added new ISA features including ALU immediates, richer addressing modes and stack-based function calls, allowing for more expressive single-cycle programs.
- Updated the CPU hardware with extra registers, multiplexers and control logic, balancing complexity and performance while testing it through assembly and Digital.

Network Security Team Project (Python, Sockets)

- Implemented both secure and subverted client-server authentication protocols using socket programming, designing a hidden backdoor that met strict requirements through group brainstorming, peer testing, and maintaining secrecy from other teams.
- Analysed 20 authentication schemes to determine if secure, identifying backdoors by spotting subtle flaws in logic and applying creative problem-solving.

Certifications

- Microsoft Certified: Azure Fundamentals (AZ-900) – Jan 2026

Extracurricular Activities

Track Captain, University Athletics Club

- Organised and led weekly track sessions for multiple training groups, coordinating with the committee and adapting sessions based on feedback.